Babu Banarasi Das University



NO SQL PROJECT ON

Student Record Management System

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OBJECTIVES:

- To understand the concept of NoSQL databases.
- To create a database and collection in MongoDB.
- To insert and manage 10 student records.
- To perform CRUD operations — Create, Read, Update, and Delete.
- To use comparison operators such as `$gt, $gte, $lt, $lte, $ne,` and `$in`.
- To perform sorting and filtering of student records.
- To perform search and retrieval operations based on specific conditions.
- To understand and implement AND (`$and`) and OR (`$or`) operations.
- To understand how MongoDB can be used to efficiently manage and retrieve student information.

1. CREATE DATABASE

MongoDB Query:

- *use AcademicDB*



2. CREATE COLLECTION

MongoDB Query:

- *db.createCollection("students")*

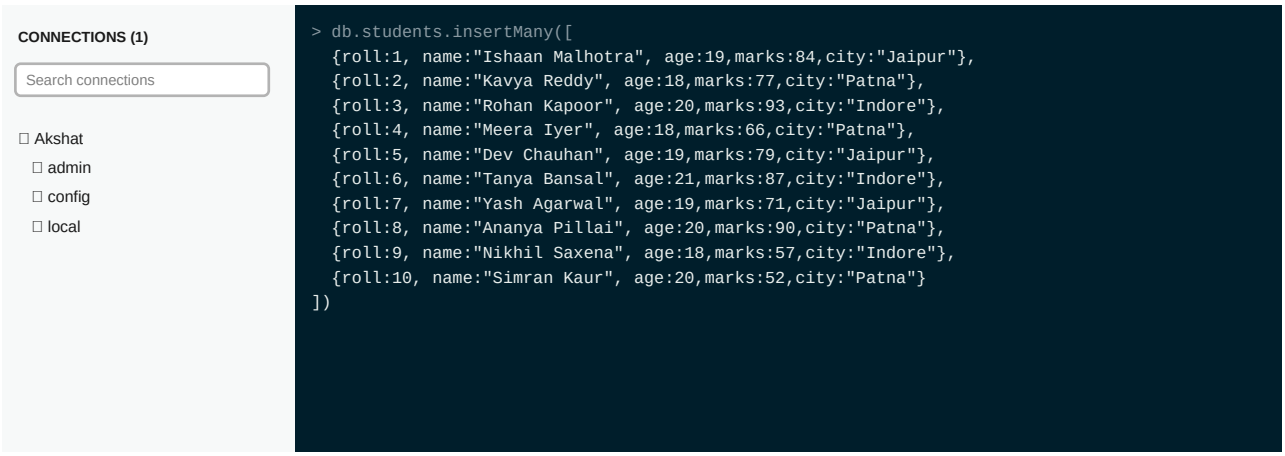
```
> db.createCollection("students")
< { ok: 1 }
```

3. INSERT 10 STUDENTS

MongoDB Query:

- *db.students.insertMany([ ])*

```
db.students.insertMany([
{roll:1, name:"Ishaan Malhotra", age:19,marks:84,city:"Jaipur"},
{roll:2, name:"Kavya Reddy", age:18,marks:77,city:"Patna"},
{roll:3, name:"Rohan Kapoor", age:20,marks:93,city:"Indore"},
{roll:4, name:"Meera Iyer", age:18,marks:66,city:"Patna"},
{roll:5, name:"Dev Chauhan", age:19,marks:79,city:"Jaipur"},
{roll:6, name:"Tanya Bansal", age:21,marks:87,city:"Indore"},
{roll:7, name:"Yash Agarwal", age:19,marks:71,city:"Jaipur"},
{roll:8, name:"Ananya Pillai", age:20,marks:90,city:"Patna"},
{roll:9, name:"Nikhil Saxena", age:18,marks:57,city:"Indore"},
{roll:10, name:"Simran Kaur", age:20,marks:52,city:"Patna"}
])
```


CONNECTIONS (1)
Search connections
Akshat
admin
config
local
> db.students.insertMany([
{roll:1, name:"Ishaan Malhotra", age:19,marks:84,city:"Jaipur"},
{roll:2, name:"Kavya Reddy", age:18,marks:77,city:"Patna"},
{roll:3, name:"Rohan Kapoor", age:20,marks:93,city:"Indore"},
{roll:4, name:"Meera Iyer", age:18,marks:66,city:"Patna"},
{roll:5, name:"Dev Chauhan", age:19,marks:79,city:"Jaipur"},
{roll:6, name:"Tanya Bansal", age:21,marks:87,city:"Indore"},
{roll:7, name:"Yash Agarwal", age:19,marks:71,city:"Jaipur"},
{roll:8, name:"Ananya Pillai", age:20,marks:90,city:"Patna"},
{roll:9, name:"Nikhil Saxena", age:18,marks:57,city:"Indore"},
{roll:10, name:"Simran Kaur", age:20,marks:52,city:"Patna"}
])

4. DISPLAY ALL STUDENTS

MongoDB Query:

- *db.students.find()*

```
> db.students.find()
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2000'),
    roll: 1,
    name: 'Ishaan Malhotra',
    age: 19,
    marks: 84,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2001'),
    roll: 2,
    name: 'Kavya Reddy',
    age: 18,
    marks: 77,
    city: 'Patna'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2002'),
    roll: 3,
    name: 'Rohan Kapoor',
    age: 20,
    marks: 93,
    city: 'Indore'
  }
```

5. EQUAL TO — $eq

MongoDB Query:

- *db.students.find({marks: {$eq: 77}})*

```
> db.students.find({marks: {$eq: 77}})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2001'),
    roll: 2,
    name: 'Kavya Reddy',
    age: 18,
    marks: 77,
    city: 'Patna'
  }
```

6. GREATER THAN — $gt

MongoDB Query:

- *db.students.find({marks: {$gt: 60}})*

Explanation: Finds students whose marks are greater than 60.

```
> db.students.find({marks: {$gt: 60}})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2000'),
    roll: 1,
    name: 'Ishaan Malhotra',
    age: 19,
    marks: 84,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2001'),
    roll: 2,
    name: 'Kavya Reddy',
    age: 18,
    marks: 77,
    city: 'Patna'
  }
```

7. LESS THAN — $lt

MongoDB Query:

- *db.students.find({marks: {$lt: 60}})*

```
> db.students.find({marks: {$lt: 60}})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2008'),
    roll: 9,
    name: 'Nikhil Saxena',
    age: 18,
    marks: 57,
    city: 'Indore'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2009'),
    roll: 10,
    name: 'Simran Kaur',
    age: 20,
    marks: 52,
    city: 'Patna'
  }
AcademicDB >
```

8. GREATER THAN OR EQUAL TO — $gte

MongoDB Query:

- *db.students.find({marks: {$gte: 90}})*

```
> db.students.find({marks: {$gte: 90}})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2002'),
    roll: 3,
    name: 'Rohan Kapoor',
    age: 20,
    marks: 93,
    city: 'Indore'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2007'),
    roll: 8,
    name: 'Ananya Pillai',
    age: 20,
    marks: 90,
    city: 'Patna'
  }
```

9. LESS THAN OR EQUAL TO — $lte

MongoDB Query:

- *db.students.find({marks: {$lte: 70}})*

```
> db.students.find({marks: {$lte: 70}})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2003'),
    roll: 4,
    name: 'Meera Iyer',
    age: 18,
    marks: 66,
    city: 'Patna'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2008'),
    roll: 9,
    name: 'Nikhil Saxena',
    age: 18,
    marks: 57,
    city: 'Indore'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2009'),
    roll: 10,
    name: 'Simran Kaur',
    age: 20,
    marks: 52,
    city: 'Patna'
  }
```

10. UPDATE OPERATION

MongoDB Query:

- *db.students.updateOne(*
  {roll:4},
  {$set:{marks:74}}
)

```
> db.students.updateOne({roll: 4}, {$set: {marks: 74}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
AcademicDB >
```

```
> db.students.find({roll: 4})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2003'),
    roll: 4,
    name: 'Meera Iyer',
    age: 18,
    marks: 74,
    city: 'Patna'
  }
AcademicDB >
```

11. DELETE ONE OPERATION

MongoDB Query:

- *db.students.deleteOne({roll:10})*

```
> db.students.deleteOne({roll: 10})
< {
    acknowledged: true,
    deletedCount: 1
  }
AcademicDB >
```

12. DELETE MANY OPERATION

MongoDB Query:

db.students.deleteMany({
 marks: {$lt:60}
})

```
> db.students.deleteMany({marks: {$lt: 60}})
< {
    acknowledged: true,
    deletedCount: 1
  }
AcademicDB >
```

13. AND OPERATION — $and

MongoDB Query:

- *db.students.find({*
  $and:[
  {marks:{$gt:80}},
  {age:21}
  ]
  })

```
> db.students.find({ $and: [ {marks: {$gt: 80}}, {age: 21} ] })
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2005'),
    roll: 6,
    name: 'Tanya Bansal',
    age: 21,
    marks: 87,
    city: 'Indore'
  }
```

14. OR OPERATION — $or

MongoDB Query:

- *db.students.find({*
  $or:[
  {city:"Jaipur"},
  {city:"Indore"}]})

```
> db.students.find({ $or: [ {city: "Jaipur"}, {city: "Indore"} ] })
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2000'),
    roll: 1,
    name: 'Ishaan Malhotra',
    age: 19,
    marks: 84,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2002'),
    roll: 3,
    name: 'Rohan Kapoor',
    age: 20,
    marks: 93,
    city: 'Indore'
  }
```

15. SORT OPERATION

MongoDB Ascending order Query:

- *db.students.find().sort({marks : 1})*

```
> db.students.find().sort({marks: 1})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2006'),
    roll: 7,
    name: 'Yash Agarwal',
    age: 19,
    marks: 71,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2003'),
    roll: 4,
    name: 'Meera Iyer',
    age: 18,
    marks: 74,
    city: 'Patna'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2001'),
    roll: 2,
    name: 'Kavya Reddy',
    age: 18,
    marks: 77,
    city: 'Patna'
  }
```

B. MongoDB Descending Order Query:

- *db.students.find().sort({marks: -1})*

```
> db.students.find().sort({marks: -1})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2002'),
    roll: 3,
    name: 'Rohan Kapoor',
    age: 20,
    marks: 93,
    city: 'Indore'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2007'),
    roll: 8,
    name: 'Ananya Pillai',
    age: 20,
    marks: 90,
    city: 'Patna'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2005'),
    roll: 6,
    name: 'Tanya Bansal',
    age: 21,
    marks: 87,
    city: 'Indore'
  }
```

16. LIMIT OPERATION

MongoDB Query:

- *db.students.find().limit(5)*

```
> db.students.find().limit(5)
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2000'),
    roll: 1,
    name: 'Ishaan Malhotra',
    age: 19,
    marks: 84,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2001'),
    roll: 2,
    name: 'Kavya Reddy',
    age: 18,
    marks: 77,
    city: 'Patna'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2002'),
    roll: 3,
    name: 'Rohan Kapoor',
    age: 20,
    marks: 93,
    city: 'Indore'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2003'),
    roll: 4,
    name: 'Meera Iyer',
    age: 18,
    marks: 74,
    city: 'Patna'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2004'),
    roll: 5,
    name: 'Dev Chauhan',
    age: 19,
    marks: 79,
    city: 'Jaipur'
  }
```

17. SEARCH BY CITY

MongoDB Query:

- *db.students.find({city:"Jaipur"})*

```
> db.students.find({city:"Jaipur"})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2000'),
    roll: 1,
    name: 'Ishaan Malhotra',
    age: 19,
    marks: 84,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2004'),
    roll: 5,
    name: 'Dev Chauhan',
    age: 19,
    marks: 79,
    city: 'Jaipur'
  }
  {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2006'),
    roll: 7,
    name: 'Yash Agarwal',
    age: 19,
    marks: 71,
    city: 'Jaipur'
  }
```

18. COUNT DOCUMENTS

MongoDB Query:

- *db.students.countDocuments()*

```
> db.students.countDocuments()
< 8
```

19.UPDATE MANY OPERATION

MongoDB Query:

- *db.students.updateMany({city: "Jaipur"}, {$set: {state: "RJ"}})*

```
> db.students.updateMany({city: "Jaipur"}, {$set: {state: "RJ"}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
```

```
> db.students.findOne({roll: 1})
< {
    _id: ObjectId('6b2f4a91c3d7e85b0a6f2000'),
    roll: 1,
    name: 'Ishaan Malhotra',
    age: 19,
    marks: 84,
    city: 'Jaipur',
    state: 'RJ'
  }
```

20.DISTINCT

MongoDB Query:

- *db.students.distinct("city")*

```
> db.students.distinct("city")
< [
    'Indore',  'Jaipur',
    'Patna'
  ]
```